

# PDN Master Formula & Plug-in Reference

## Module 0: Performance

### Question Type: Calculating Speedup

**Formula:**  $S = T_{serial}/T_{parallel}$

**Example Problem:** Suppose a serial plant takes  $10^6$ s and a parallel plant takes  $2 \times 10^3$ s. What is the speedup?

**Calculation:**  $S = 1,000,000/2,000 = 500$ .

### Question Type: Calculating Efficiency

**Formula:**  $E = S/p$

**Example Problem:** With a speedup of 500 and using 1024 processors, what is the efficiency?

**Calculation:**  $E = 500/1024 \approx 0.49$  (49%).

### Question Type: Amdahl's Law

**Formula:**  $S_{max} = 1/[(1 - f) + f/p]$

**Example Problem:** If 90% of code is parallel, what is the max speedup?

**Calculation:**  $f = 0.9 \rightarrow S_{max} = 1/(1 - 0.9) = 10\times$ .

## Module 1: OpenMP

### Question Type: Trapezoidal Rule Integration

**Formula:**  $h = (b - a)/n$ ,

Area  $\approx \frac{h}{2}[f(a) + f(b) + 2 \sum f(x_i)]$

**Example Problem:** Integrate  $f(x)$  from 0 to 10 using 5 segments.

**Calculation:**  $h = 10/5 = 2$ .  $x_i = \{0, 2, 4, 6, 8, 10\}$ .

Area  $= \frac{2}{2} \times [f(0) + f(10) + 2(f(2) + f(4) + f(6) + f(8))]$ .

## Module 2: CUDA & GPUs

### Question Type: Execution Configuration

**Concept:** Specifies the grid and block dimensions for a launch.

**Example:** What does `<<<G, B>>>` specify? **Answer:** The grid and block dimensions.

### Question Type: Grid and Block Dimensions

**Formula:**  $G = \lceil N/B \rceil$ . Total threads  $= G \times B$ .

**Example Problem:**  $N = 2000$ ,  $B = 256$ . How many blocks?

**Calculation:**  $\lceil 2000/256 \rceil = 8$  blocks. Total threads  $= 2048$ .

### Question Type: Global Coordinate Calculation

**Formula:**

$GlobalX = blockIdx.x \times blockDim.x + threadIdx.x$

**Example:**

$blockIdx.x = 4, threadIdx.x = 11, blockDim.x = 32$ .

What is Global X?

**Calculation:**  $4 \times 32 + 11 = 128 + 11 = 139$ .

### Question Type: Local Thread Index from Global

**Formula:**  $threadIdx.x = GlobalCol \pmod{blockDim.x}$

**Example:** Pixel at  $[col = 91, row = 130]$ , Block  $16 \times 16$ .

**Calculation:**  $91 \pmod{16} = 11$ .  $130 \pmod{16} = 2$ .

**Result:**  $threadIdx.x = 11, threadIdx.y = 2$ .

### Question Type: Boundary Conditions

**Concept:** Grid size often exceeds actual data size.

**Example:** Why are boundary checks needed? **Answer:**  
The grid may contain more threads than pixels.

### **Question Type: Stencil Special Handling**

**Concept:** Neighbor pixels may fall outside image.

**Example:** Why do stencils need special handling?

**Answer:** Locations may fall outside the valid input domain.

## **Module 5: Network**

### **Question Type: Transmission Delay**

**Formula:**  $d_{trans} = L/R$

**Example Problem:** Packet is 40,000 bits, link is 4 Mbps.

**Calculation:**  $40,000/4,000,000 = 10\text{ms}$ .

### **Question Type: Propagation Delay**

**Formula:**  $d_{prop} = d/s$

**Example Problem:** 5,000 km, speed  $2.5 \times 10^8$  m/s.

**Calculation:**  $5 \times 10^6/2.5 \times 10^8 = 20\text{ms}$ .